



Point of View

The role of blood-based biomarkers in managing the epidemic of Alzheimer's disease in India

Rishab Gupta¹

¹Department of Psychiatry, Brigham and Women's Faulkner Hospital, Boston, Massachusetts, Unites States

***Corresponding author:**

Dr. Rishab Gupta,
Department of Psychiatry,
Brigham and Women's
Faulkner Hospital, Boston,
Massachusetts, Unites States.

rishabaiims@gmail.com

Received: 26 August 2025

Accepted: 27 September 2025

Epub Ahead of Print:

07 January 2026

Published: ***

DOI:

10.25259/FH_77_2025

Quick Response Code:



ABSTRACT

Despite being the most common cause of dementia in the world, Alzheimer's disease (AD) commonly remains underdiagnosed. The burden of AD is felt the most by the developing world and has been rising steadily with the increase in population longevity and increasing incidence of metabolic and vascular risk factors. Several advances have been made in understanding the neurobiological underpinnings of AD. AD is now diagnosed based on its biology, utilizing specific biomarkers as opposed to following the clinical approach, which is less precise. However, the currently used biomarkers (cerebrospinal fluid (CSF) biomarkers and Amyloid positron emission tomography (PET) scan findings) pose a huge challenge in terms of their global adoption because of their low affordability, poor accessibility, and perceived invasiveness. This challenge serves as a major roadblock in the timely and accurate diagnosis of AD, its prognostication, and in terms of pursuing newer targeted therapeutic approaches, which slow down the disease trajectory to some extent. As a result, blood-based biomarkers (BBMs) of AD are being developed. These have an immense potential to transform the landscape of AD diagnosis, treatment, prognostication, and research. Low- and middle-income countries like India should reflect on the role of BBMs of AD and consider ways to either develop them indigenously or adopt them in their healthcare systems to manage the worsening epidemic of AD. This paper discusses the opportunities and challenges faced by India in the context of utilizing BBMs of AD.

Keywords: Alzheimer's disease, Blood-based biomarkers, Plasma markers

Alzheimer's disease (AD) is the most common cause of neurodegenerative dementia worldwide, responsible for 60-70% of the total cases of dementia overall.¹ In 2021, the global prevalence of dementia was ~57 million cases, with a substantial proportion occurring in low- and middle-income countries (LMICs) due to population expansion and aging.¹ As per the World Health Organization (WHO) Global Burden of Disease (GBD) study, the burden of AD and other dementias has been rising in absolute numbers in LMICs like India in the last three decades.² The age-standardized rates of AD and other dementias are also increasing in India, albeit at a slower pace compared to developed nations.² The Harmonized Diagnostic Assessment of Dementia for the Longitudinal Aging Study in India (LASI-DAD) calculated that the prevalences of mild and major neurocognitive disorder in those aged 60 and above were 17.6% and 7.2% (estimated to be 9.9 million individuals), respectively.³ For the uninitiated, mild neurocognitive disorder is clinically equivalent to Mild Cognitive Impairment (MCI), and major neurocognitive disorder is synonymous with dementia. The above estimates are very concerning, given that in India, the proportion of people above age 60 is projected to comprise 20% of the population by 2050.³ Dementia is the seventh leading cause of death in the world, and a major cause of disability.¹ As per the WHO GBD 2021 study, the AD and other dementias-related deaths in India quadrupled

from 1990 to 2021.⁴ In India, the projected mortality because of AD and other dementias will keep increasing until at least 2050.⁴

Worldwide, the rate of undetected dementia was estimated to be 61.7%.⁵ The situation in India is no different. A recent study from India found that >40% of patients with dementia were diagnosed >2 years after the symptom onset in a memory disorders clinic in a major metropolitan city.⁶ Additionally, Indian society suffers from poor awareness regarding dementia. Not only lay people but even health professionals lack knowledge and suffer from dementia-related misconceptions and stigma.^{7,8} India does not have a dementia-specific national program. That said, dementia is being addressed through other programs: the National Programme for Prevention and Control of Cancer, Diabetes, Cardiovascular Diseases and Stroke (NPCDCS), the National Programme for Health Care of the Elderly (NPHCE), and the central and state mental health institutions and the psychiatric wings of the central and state medical colleges.⁹

Despite the above promising initiatives, there is an urgent need for improved awareness, targeted prevention, and control strategies to check the impact of AD and other dementias on the Indian population.

In India, the International Classification of Diseases (ICD) is commonly used to diagnose various neuropsychiatric disorders. According to the ICD-11, AD is described as a progressive neurodegenerative disease characterized by memory impairment, decline in other cognitive domains, and functional deterioration. Unfortunately, the ICD-11 does not specify operational diagnostic criteria or biomarker requirements for the AD diagnosis.¹⁰ This is a significant limitation considering that the diagnostic accuracy of a clinical diagnosis of AD, even among experts, is not acceptably high.¹¹ Clinical diagnosis of AD (without using the specific biomarkers) misclassifies the dementia subtype in up to 35% of cases, even in specialized clinics, with higher rates in primary care settings.¹² This happens because dementia clinical syndromes often overlap and may be caused by non-AD pathologies, leading to both false positives and false negatives.¹³ Additionally, without specific AD biomarkers, it becomes difficult for clinicians to reliably distinguish AD from other neurodegenerative dementias (like Frontotemporal dementia, vascular dementia, etc.), especially in atypical presentations or early disease stages.¹⁴ This hampers early diagnosis, which is critical for timely intervention, especially since initiation of anti-amyloid immunotherapy requires biomarker confirmation. In India, AD is diagnosed clinically without the help of any specific biomarkers.¹⁵ Even Magnetic resonance imaging (MRI) brain, the commonly used neuroimaging modality in the AD work-up, is not readily

available or sought due to financial reasons in India.^{16,17} These are massive barriers in the path to diagnosing AD precisely. Without specific AD biomarkers, the diagnostic work-up and treatment of AD suffer significantly.

The latest diagnostic criteria laid down by the Alzheimer's Association workgroup state that an abnormal core 1 biomarker (amyloid positron emission tomography (PET), certain cerebrospinal fluid (CSF) biomarkers, and specific blood-based biomarkers (BBMs)) is sufficient to diagnose AD even when a patient is clinically asymptomatic.¹⁸ These core 1 biomarkers tend to become abnormal early in the disease process and indicate the presence of AD-related neuropathologic changes (amyloid plaques and tau tangles). The core 2 biomarkers (certain soluble tau fragments associated with tau protein deposition in the brain and tau PET) become abnormal later in the disease trajectory, have prognostic value, and also suggest that AD is contributing to clinical symptoms.¹⁸

As one may imagine, the discovery and clinical implementation of AD biomarkers is among the most exciting developments in the field of AD. Amyloid PET scan can demonstrate (both qualitatively and quantitatively) the abnormal deposition of amyloid protein in the relevant brain regions, quite early in the disease process. In a study, it was found that the Amyloid PET scan helped change the diagnosis from AD to non-AD in 25% of cases and informed clinical management in nearly two-thirds of all patients.¹⁹

Compared to developed countries, which have the privilege of a relatively abundant trained clinical workforce, infrastructure, and financial means to diagnose and manage AD, LMICs face unique challenges in managing AD. Some of these are as follows:

1. There is a scarcity of trained clinicians who can accurately diagnose and manage AD.¹⁵
2. Testing for AD biomarkers (Amyloid PET, CSF AD biomarkers, and blood-based biomarkers) is currently unavailable in India. In India, the diagnosis of AD is still made clinically, with the help of typical history, physical examination, and cognitive evaluation.¹⁵
3. Even if amyloid PET and CSF AD biomarkers are made available in India, it would still be quite challenging to use them clinically, simply because of their very high cost (in the US, an Amyloid PET scan costs up to \$5000, whereas obtaining the CSF AD biomarkers would cost up to \$1000).^{20,21}
4. While there are no data on the acceptance of Lumbar puncture in India, the studies from other developing countries (including India's neighbor, Iran) have clearly shown a very negative attitude and skepticism towards

Lumbar puncture in general.^{22,23} It would not be surprising to find similar results if such a survey were done in India.

5. Additionally, even if the cost of the above biomarkers is brought down somehow, the lack of PET scanners, scarcity of trained technicians, and clinicians who can interpret the PET scans, and perform lumbar puncture to obtain CSF, is difficult to overcome.²⁴

In light of the above barriers and considering the relative ease of obtaining the testing and gentler learning curve of the test interpretation compared to the Amyloid PET and CSF AD biomarkers, the BBMs of AD are conceivably the easiest alternative that could be developed indigenously, or “imported,” adopted, and scaled up in India.

Several BBMs of AD have been developed, including but not limited to Amyloid β 42/40 ratio, p-tau217, p-tau181, p-tau231, neurofilament light chain (NFL), and Glial Fibrillary Acidic Protein (GFAP).²⁵ Apart from being used in clinical practice by some clinicians, BBMs are increasingly being utilized in clinical trials to speed up patient enrollment, to reduce costs, and to assess the efficacy of disease-modifying treatments. In May 2025, the US Food and Drug Administration approved the first blood test for diagnosing AD among patients who are ≥ 55 years and exhibit clinical features of the disease, the Lumipulse G pTau217/ β -Amyloid-42 Plasma Ratio.²⁶ This test measures two plasma proteins, p-tau217 and A β 42, and calculates their numerical ratio, which correlates with the presence or absence of amyloid plaques in the brain. This approval was granted on the heels of a multi-center study that showed that 91.7% of individuals with the positive plasma biomarker had positive amyloid plaques detected on the amyloid PET scan or CSF AD biomarker test result, and 97.3% of individuals with negative plasma biomarker results had a negative amyloid PET scan or CSF AD biomarker test result. It is crucial to note that this plasma biomarker should not be used as a screening test or as a stand-alone diagnostic test currently.

Alzheimer's Association's Clinical Practice Guidelines recommend using only those BBMs that have sensitivity and specificity $\geq 90\%$ as a replacement for Amyloid PET imaging or CSF AD biomarker testing in patients with cognitive impairment who present to the specialized care settings for memory disorders. On the other hand, those BBMs with $\geq 90\%$ sensitivity and $\geq 75\%$ specificity can be used for triaging.²⁷

There are three main arguments against the proposal of adopting BBMs of AD in India. First, the BBMs have not been universally validated.²⁸ Most of the validation studies were carried out in White populations. BBMs of AD do not have agreed-upon cut-offs; their sensitivities and specificities vary, their performance is context-of-use dependent, and

they are prone to interference from certain medications and medical illnesses.²⁵ These have not yet been shown to directly improve patient outcomes like slowed disease progression and improved quality of life. Secondly, the BBMs of AD are also not quite affordable currently, costing anywhere from a few hundred to around one thousand dollars. Thirdly, even if AD is diagnosed with the help of plasma biomarkers, disease-modifying monoclonal antibodies against AD (Lecanemab and Donanemab) are not very effective,²⁹ and are still unavailable in India, and even when they become available, their cost is likely to be quite prohibitive.

The issue of non-affordability of BBMs can be resolved. There are three approaches by which it can be achieved: First, by indigenously developing accurate BBMs of AD with the help of local research centers or via public-private partnership. Secondly, by forging collaborations with research centers in developed countries, like the US. Thirdly, by exploring the purchase of test assays from the organizations that developed the tests and then scaling up the operation over time. Bulk ordering could bring the cost of the assay down for individual patients.

Each of the proposed approaches has some flaws. Minimal investment in healthcare research poses a major challenge in developing sophisticated test assays for BBMs of AD in India.³⁰ In the development stage, the blood-based test assays in the US were typically validated against the gold-standard diagnosis of AD (requiring expensive biomarkers, i.e., Amyloid PET scan or CSF AD biomarkers),³¹ which are not available in India. Therefore, it would be a humongous task to develop BBMs of AD in India. Indian medical research institutions could collaborate with relevant research centers in countries like the US by exploring like-minded individuals/departments, but even if that happened, scaling up the process would require a deep interest, strong will, and massive support from funding bodies in India. The counter to the third argument against the adoption of BBMs of AD in India, i.e., monoclonal antibodies against AD being insufficiently effective, unavailable in India, and likely to be extremely expensive, is that a more accurate AD diagnosis can at least help patients and their families plan better for the future.²⁵ Living with the fear of having AD and its deteriorating course can exert a massive mental toll on patients and their families. As one could imagine, differentiating a non-progressive or potentially reversible cognitive decline from slowly progressive AD can be very reassuring for the patients and their caregivers. It will also help make a better estimation of the burden of AD, which can then inform policymakers in strategy planning to tackle the AD epidemic in India. These biomarkers can also help differentiate AD from other progressive neurodegenerative diseases like Frontotemporal Lobar Degeneration.³² Even if current AD treatments are not effective and are unavailable, the future of

AD therapeutics is quite promising. Drugs against tau protein deposition, active and passive immunotherapies, antisense oligonucleotides, gene therapies, and nanoparticle-based drug delivery systems, among others, are being explored.³³ Additionally, several non-pharmacologic interventions like cognitive retraining, exercise, and social engagement can still be employed in the absence of disease-modifying drugs.³⁴

To sum up, presently, we are in an era of great optimism for AD stakeholders, given that for the first time in the history of the disease, we have access to disease-modifying treatments and multiple, highly accurate biomarkers of the disease. The BBMs of AD have an immense potential to reduce the global inequity seen in the diagnosis and management of AD. Unfortunately, the BBMs of AD are not yet ready for use in the primary care clinics where most of the patients with cognitive disorders are managed in LMICs like India. However, their judicious use in the specialized memory clinics is now happening in the developed world. LMICs like India must not remain passive in the development/adoption of these valuable testing modalities.

Acknowledgment: I would like to thank my dear colleague, Dr Snehil Gupta, for supporting this topic. Additionally, Drs. Kirk Daffner, Scott McGinnis, and Seth Gale deserve my heartfelt gratitude for demystifying the complexity of AD for me.

Author contribution: RG: Concept and design of the study, interpretation of the literature cited, drafting of the article, its critical revision, and final approval of the version to be published. RG is accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Ethical approval: Institutional Review Board approval is not required.

Declaration of patient consent: Patient's consent not required as there are no patients in this study.

Financial support and sponsorship: Nil.

Conflicts of interest: There are no conflicts of interest.

Use of artificial intelligence (AI)-assisted technology for manuscript preparation: The authors confirm that there was no use of Artificial Intelligence (AI)-Assisted Technology for assisting in the writing or editing of the manuscript and no images were manipulated using AI.

REFERENCES

1. Dementia. Available from <https://www.who.int/news-room/fact-sheets/detail/dementia>. [Last accessed on 2025 Aug 12].
2. Yin M, Ding S, Zhao M, Liu J, Su W, Fu M, *et al.* The global, regional, and national burden and attributable risk factors of Alzheimer's disease and other dementias, 1990-2021: A systematic analysis for the Global Burden of Disease Study. *J Alzheimer's Dis* 2025;107:192-206.
3. Gross AL, Nichols E, Angrisani M, Ganguli M, Jin H, Khobragade P, *et al.* Prevalence of DSM-5 mild and major neurocognitive disorder in India: Results from the LASI-DAD. *PLoS One* 2024;19:e0297220.
4. Fang H, Li L, Ai J, Wang Q. Mortality from Alzheimer's disease and other dementias and its heterogeneity across states in India findings from GBD 2021 study. *Int J Surg* 2025;111:4830-41.
5. Lang L, Clifford A, Wei L, Zhang D, Leung D, Augustine G, *et al.* Prevalence and determinants of undetected dementia in the community: A systematic literature review and a meta-analysis. *BMJ Open* 2017;7:e011146.
6. Ellajosyula R, Narayanan J, Hegde S, Kamath V, Murgod U, Easwaran V, *et al.* Delay in the diagnosis of dementia in urban India: Role of dementia subtype and age at onset. *Int J Geriatr Psychiatry* 2022;37.
7. Hurzuk S, Farina N, Pattabiraman M, Ramasamy N, Alladi S, Rajagopalan J, *et al.* Understanding, experiences and attitudes of dementia in India: A qualitative study. *Dementia (London)* 2022;21:2288-306.
8. Marathe A, Dharaiya D, Gohel M, Prabhakaran A, Varma J. Knowledge about Alzheimer's disease among physicians: A cross-sectional study from Anand, Gujarat. *J Geriatr Ment Health* 2023;10:77-80.
9. Efforts by the Government to reduce Dementia cases, address mental disorders and curb the use of Tobacco. Available from <https://www.pib.gov.in/www.pib.gov.in/Pressreleaseshare.aspx?PRID=1796437>. [Last accessed on 2025 Aug 12].
10. Gaebel W, Jessen F, Kanba S. Neurocognitive disorders in ICD-11: The debate and its outcome. *World Psychiatry* 2018;17:229-30.
11. Beach TG, Monsell SE, Phillips LE, Kukull W. Accuracy of the clinical diagnosis of alzheimer disease at National institute on aging Alzheimer disease centers, 2005-2010. *J Neuropathol Exp Neurol* 2012;71:266-73.
12. Palmqvist S, Tideman P, Mattsson-Carlsson N, Schindler SE, Smith R, Ossenkoppele R, *et al.* Blood biomarkers to detect Alzheimer disease in primary care and secondary care. *JAMA* 2024;332:1245-57.
13. Bouteloup V, Villain N, Vidal JS, Gonzalez-Ortiz F, Yuksekeli I, Santos C, *et al.* Cognitive phenotyping and interpretation of Alzheimer blood biomarkers. *JAMA Neurol* 2025;82:506-15.
14. Dubois B, Villain N, Schneider L, Fox N, Campbell N, Galasko D, *et al.* Alzheimer disease as a clinical-biological construct-an international working group recommendation. *JAMA Neurol* 2024;81:1304-11.
15. Babulal GM, Zha W, Trani J-F, Guerra JL, Tee BL, Zhu Y, *et al.* Identifying gaps and barriers in Alzheimer's disease and related dementia research and management in low- and middle-income countries: A survey of health professionals and researchers. *JAD* 2024;101:1307-20.
16. Ragavan D. Commentary - radiology in India: The next decade. *Indian J Radiol Imaging* 2008;18:187-9.
17. Jalloul M, Miranda-Schaeubinger M, Noor AM, Stein JM, Amiruddin R, Derbew HM, *et al.* MRI scarcity in low- and middle-income countries. *NMR Biomed* 2023;36:e5022.
18. Jack CR, Andrews JS, Beach TG, Buracchio T, Dunn B, Graf A, *et al.* Revised criteria for diagnosis and staging of Alzheimer's disease: Alzheimer's association workgroup. *Alzheimer's & Dementia* 2024;20:5143-69.
19. Rabinovici GD, Gatsonis C, Apgar C, Chaudhary K, Gareen I, Hanna L, *et al.* Association of amyloid positron emission tomography with subsequent change in clinical management

- among medicare beneficiaries with mild cognitive impairment or dementia. *JAMA* 2019;321:1286-94.
20. Mandal PK, Maroon JC, Garg A, Arora NK, Bansal R, Kaushik A, *et al.* Blood biomarkers in Alzheimer's disease. *ACS Chem Neurosci* 2023;14:3975-8.
 21. Dhauria M, Mondal R, Deb S, Shome G, Chowdhury D, Sarkar S, *et al.* Blood-based biomarkers in Alzheimer's disease: Advancing non-invasive diagnostics and prognostics. *IJMS* 2024;25:10911.
 22. Borhani-Haghighi A, Rezaei R, Etemadi S, Ghaem H, Shariat A. Knowledge and attitudes of Iranian patients with regard to lumbar puncture. *Neurosciences (Riyadh)* 2009;14:360-3.
 23. Aldayel AY, Alharbi MM, Almasri MS, Alkhonezan SM. Public knowledge and attitude toward lumbar puncture among adults in Riyadh, Saudi Arabia: A cross-sectional study. *SAGE Open Med* 2019;7:2050312119871066.
 24. Páez D, Orellana P, Gutiérrez C, Ramirez R, Mut F, Torres L. Current status of nuclear medicine practice in Latin America and the Caribbean. *J Nucl Med* 2015;56:1629-34.
 25. Hampel H, Hu Y, Cummings J, Mattke S, Iwatsubo T, Nakamura A, *et al.* Blood-based biomarkers for Alzheimer's disease: Current state and future use in a transformed global healthcare landscape. *Neuron* 2023;111:2781-99.
 26. Commissioner O of the. FDA clears first blood test used in diagnosing Alzheimer's disease. FDA. May 28, 2025. <https://www.fda.gov/news-events/press-announcements/fda-clears-first-blood-test-used-diagnosing-alzheimers-disease>. Accessed August 25, 2025.
 27. Palmqvist S, Whitson HE, Allen LA, Suarez-Calvet M, Galasko D, Karikari TK, *et al.* Alzheimer's association clinical practice guideline on the use of blood-based biomarkers in the diagnostic workup of suspected Alzheimer's disease within specialized care settings. *Alzheimer's & Dementia* 2025;21.
 28. Lista S, Mapstone M, Caraci F, Emanuele E, López-Ortiz S, Martín-Hernández J, *et al.* A critical appraisal of blood-based biomarkers for Alzheimer's disease. *Ageing Res Rev* 2024;96:102290.
 29. Qiao Y, Gu J, Yu M, Chi Y, Ma Y. Comparative Efficacy and safety of monoclonal antibodies for cognitive decline in patients with Alzheimer's disease: A systematic review and network meta-analysis. *CNS Drugs* 2024;38:169-92.
 30. Ahuja M, Pradeep Y. Medical research in India: Let us plug the holes! *J Midlife Health* 2023;14:63-5.
 31. Zhong X, Wang Q, Yang M, Lin G, Yao K, Wu Z, *et al.* Plasma p-tau217 and p-tau217/A β 1-42 are effective biomarkers for identifying CSF- and PET imaging-diagnosed Alzheimer's disease: Insights for research and clinical practice. *Alzheimer's & Dementia* 2025;21.
 32. VandeVrede L, Cho H, Sanderson-Cimino M, Wekselman F, Cobigo Y, Gorno-Tempini ML, *et al.* Detection of Alzheimer neuropathology in Alzheimer and Non-Alzheimer clinical syndromes with blood-based biomarkers. *JAMA Neurol* 2025;82:344-5.
 33. Ahmad F, Karan A, Sharma R, Sharma NS, Sundar V, Jayaraj R, *et al.* Evolving therapeutic interventions for the management and treatment of Alzheimer's disease. *Ageing Res Rev* 2024;95:102229.
 34. Martínez-López S, Tabone M, Clemente-Velasco S, González-Soltero MdelRío, Bailén Mía, de Lucas B, *et al.* A systematic review of lifestyle-based interventions for managing Alzheimer's disease: Insights from randomized controlled trials. *J Alzheimer's Dis* 2024;102:943-66.

How to cite this article: Gupta R. The role of blood-based biomarkers in managing the epidemic of Alzheimer's disease in India. *Future Health*. doi: 10.25259/FH_77_2025